# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** A lot of 320 ball pens contains 48 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

**(a)** $\frac{271}{320}$ **(b)** $\frac{17}{20}$ **(c)** $\frac{273}{320}$ **(d)** $\frac{3}{20}$

**2.** Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the product of the numbers is 12 is: 1

**(a)** $\frac{1}{9}$ **(b)** $\frac{5}{36}$ **(c)** $\frac{1}{12}$ **(d)** $\frac{8}{9}$

**3.** Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum is a prime number is: 1

**(a)** $\frac{7}{12}$ **(b)** $\frac{7}{18}$ **(c)** $\frac{4}{9}$ **(d)** $\frac{5}{12}$

**4.** A bag contains 9 red balls and $x$ blue balls. The probability of drawing a blue ball is 7 times the probability of drawing a red ball. The value of $x$ is: 1

**(a)** 62 **(b)** 63 **(c)** 7 **(d)** 72

**5.** A bag contains 9 red and 15 blue balls. When $x$ more red balls are added, the probability of drawing a red ball becomes $\frac{1}{2}$. The value of $x$ is: 1

**(a)** 7 **(b)** 6 **(c)** 12 **(d)** 5

**6.** A bag contains 17 red and 3 blue balls. 13 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

**(a)** $\frac{10}{11}$ **(b)** $\frac{1}{11}$ **(c)** $\frac{31}{33}$ **(d)** $\frac{29}{33}$

**7.** A box contains 90 tickets numbered 1 to 90. One ticket is drawn at random. The probability that the number drawn is a multiple of 2 but not of 4 is: 1

**(a)** $\frac{4}{15}$ **(b)** $\frac{11}{45}$ **(c)** $\frac{67}{90}$ **(d)** $\frac{23}{90}$

**8.** A lot of 400 ball pens contains 52 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

**(a)** $\frac{13}{100}$ **(b)** $\frac{349}{400}$ **(c)** $\frac{87}{100}$ **(d)** $\frac{347}{400}$

**9.** **Assertion (A):** If $P(E) = 0.35$, then $P(\text{not } E) = 0.65$. 1

**Reason (R):** For any event E, $P(E) + P(\text{not } E) = 1$.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**10.** **Assertion (A):** A bag has 3 red and 5 blue balls. The probability of drawing a red ball is $\frac{3}{8}$. 1

**Reason (R):** The probability of an event is the number of unfavourable outcomes divided by the total number of outcomes.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** A bag contains 21 red and 9 blue balls. 9 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 2

**12.** Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is 5. 2

**13.** A bag contains 12 red balls and $x$ blue balls. The probability of drawing a blue ball is 7 times the probability of drawing a red ball. Find the value of $x$. 2

**14.** A bag contains 10 red and 14 blue balls. 6 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that neither die shows a 3. 3

**16.** A bag contains 10 red and 14 blue balls. When $x$ more red balls are added, the probability of drawing a red ball becomes $\frac{9}{16}$. Find the value of $x$. 3

**17.** A box contains 54 tickets numbered 1 to 54. One ticket is drawn at random. Find the probability that the number drawn is not a multiple of 7. 3

OR

Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the product is 6. 3

**18.** A bag contains 15 red and 5 blue balls. When $x$ more red balls are added, the probability of drawing a red ball becomes $\frac{6}{7}$. Find the value of $x$. 3

OR

A box contains 90 tickets numbered 1 to 90. One ticket is drawn at random. Find the probability that the number drawn is a prime number. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** A raffle has 25 tickets numbered 1 to 25 placed in a box. One ticket is drawn at random. 5

(i) How many equally likely outcomes are there?

(ii) Find the probability that the number is a prime number.

(iii) Find the probability that the number is a multiple of 5.

(iv) Find the probability that the number is neither prime nor a multiple of 5.

**20.** Two dice are thrown together. Find the probabilities of these events (in this order): 5

(i) at least one die shows 6,

(ii) the sum is 3 or 11,

(iii) the product is 6.

OR

A bag contains 19 red and 1 blue balls. When $x$ more red balls are added, the probability of drawing a red ball becomes $\frac{40}{41}$. Find the value of $x$. 5

## END OF QUESTION PAPER — Total: 40 Marks